Fast Tree Diffusions for Probabilistic Tabular Regression via Flow Matching

Abstract

We extend Treeffuser, a score-based diffusion model for probabilistic tabular regression with gradient-boosted trees, in two directions motivated by fast sample-based probabilistic modelling. First, a combined score-side recipe—conditional-mean residualization, EDM-style preconditioning, log-sigma time sampling, and noise-level features—improves aggregate calibration after per-dataset tuning, reducing 95% coverage error from 0.023 to 0.015 while cutting sampling time 6.5× over the published baseline. Second, we replace score matching with flow matching using Gaussian probability paths, with algebraic score recovery available when stochastic sampling is needed. With a VP path and deterministic five-step Heun ODE, the flow-matching variant gives the best normalized CRPS among Treeffuser rows and is another 2.4× faster than the improved score model, but gives back some of the score model's coverage advantage. The common mechanism is better conditioning of the supervised regression problem seen by the tree ensemble. Under this ten-dataset tuning protocol, the Treeffuser variants give the best aggregate normalized CRPS and coverage metrics among the displayed rows, though gains over the published baseline are directional rather than conventionally significant ($p \approx 0.07$, paired Wilcoxon), and tuned non-diffusion baselines still win several individual datasets on raw CRPS. Stochastic ($\varepsilon > 0$) samplers do not Pareto-dominate the deterministic corner, so the empirical recommendation is the $\varepsilon = 0$ specialization when aggregate CRPS and latency are the target, though stochasticity may improve conditional tail coverage on high-uncertainty subsets.

1 INTRODUCTION

Probabilistic regression on tabular data must combine the predictive strength of gradient-boosted trees with full predictive distributions usable for calibration, decision-making, and uncertainty quantification. Treeffuser [Beltrán-Velez et al., 2024] achieves this by training LightGBM [Ke et al., 2017] regressors to estimate the score of a noise-perturbed conditional density and integrating the corresponding reverse-time stochastic differential equation (SDE) [Song et al., 2021] at inference. The framework inherits the calibration properties of score-based diffusion while avoiding neural-network density estimators.

Our notion of tractability is computational rather than likelihood-exact. The model does not provide a closed-form marginal likelihood, and predictive quantiles, intervals, and calibration summaries are Monte Carlo estimates from generated samples. The tractable object is instead the conditional inference procedure: the final FM row uses a five-step Heun ODE to generate predictive samples, Eq. (2) converts velocities to scores in closed form when stochastic sampling is desired, and training reduces to supervised tree regression.

This paper develops Treeffuser along two axes that interact with the tree-based score regressor in ways that differ from the neural-network case. *First*, on the score-training side, we evaluate four modifications motivated by the histogram-bin density structure of tree learners. *Second*, we replace score matching with flow matching using Gaussian probability paths [Lipman et al., 2023, Liu et al., 2023], which removes the indirect score-to-drift mapping that score-matching imposes on tree predictions and opens access to the stochastic-interpolant family [Albergo et al., 2023], parameterized by a non-negative stochasticity schedule $\varepsilon(t)$ that interpolates between the deterministic ODE sampler ($\varepsilon \equiv 0$) and the score-based reverse SDE.

Contributions. We make three contributions:

1. *Score-side modifications for tree-based diffusion.* An

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evaluation of conditional-mean residualization, EDM preconditioning, log-σ time sampling, and an explicit log-σ feature in the histogram-split regime of LightGBM (§2.1).

2. *Flow matching for Treeffuser with operational score recovery.* A Gaussian-path FM trainer for Treeffuser, using Eq. (2) to recover the implied conditional score when stochastic-interpolant sampling or endpoint-error analysis requires it (§2.2, Lemma 1).
3. *Regression-conditioning unification.* We identify regression conditioning for the tree ensemble—not the functional class itself—as the bottleneck behind both score- and flow-side gains, supported by Propositions 3, 1, and 2 (§4, Appendix A).

Across ten UCI probabilistic-regression benchmarks, with one fold reserved for tuning and five folds held out for evaluation, the score-side modifications give the best aggregate calibration and CRPS skill score among Treeffuser rows, while flow matching gives the best aggregate normalized CRPS and the fastest Treeffuser sampler. The shared lesson is that tree-based diffusion models depend strongly on whether the noising path, parameterization, training distribution, features, and sampler make the induced supervised problem well matched to histogram splits. Stochastic samplers ($\varepsilon > 0$) do not Pareto-dominate the deterministic ODE corner on any aggregate metric, so the empirical recommendation collapses the one-parameter family to $\varepsilon = 0$ under this evaluation protocol.

2 METHOD

2.1 SCORE-SIDE MODIFICATIONS

The published Treeffuser uses VESDE marginals with uniform t sampling, a noise-prediction target, and a feature layout $[\, y_t,\ X,\ t\,]$ given to LightGBM. We change four pieces, each motivated by tree-based regression rather than borrowed wholesale from neural diffusion. They act on distinct levers: response/input conditioning, allocation of training rows over noise levels, explicit noise-level features, and sampler efficiency.

Conditional-mean residualization. We fit a cross-validated LightGBM mean predictor $\hat{\mu}(x)$ on the training data and apply diffusion to $y - \hat{\mu}(x)$ rather than y directly. This converts a heteroscedastic conditional density into one whose mode is approximately zero and whose remaining variation is scale only, which empirically simplifies the score-regression target. We use a high-capacity configuration (no depth cap, 63 leaves, n_{est}=300, η=0.05) selected from a prior diagnostic sweep (Appendix D) and then fixed; the residualizer is not tuned jointly with the diffusion or flow hyperparameters. On large datasets the fixed residualizer can become a bottleneck once the inner model has enough data to represent the conditional mean directly (see Appendix D and Table 2).

EDM preconditioning. Following Karras et al. [2022], we predict a denoised target $D_\theta(y_t, t)$ scaled to unit variance across noise levels and reconstruct the score $s(y_t, t) = (D_\theta - y_t)/\sigma(t)^2$. This stabilizes the regression target for trees, which would otherwise need to extrapolate noise-prediction magnitudes across t.

Log-sigma time sampling. We draw $\log \sigma(t) \sim \mathcal{N}(-1.2, 1.2^2)$, clip to the SDE's achievable range, and invert. Trees bin features by data density, so the distribution of training t values directly controls where the score model spends capacity. This is a stronger lever than loss reweighting [Hang et al., 2023], which scales loss within fixed bins.

Log-sigma noise feature. The feature vector for LightGBM becomes $[\, y_t,\ X,\ t,\ \log \sigma(t)\,]$, giving the regressor an explicit noise-level signal aligned with EDM scaling.

2.2 FLOW MATCHING WITH GAUSSIAN PATHS

A Gaussian flow path interpolates data y_0 and a standard-normal prior z via

$$y_t = \alpha(t)\, y_0 + \beta(t)\, z, \quad t \in [0, 1], \tag{1}$$

with boundary conditions $\alpha(0) = 1$, $\alpha(1) = 0$, $\beta(0) = 0$, $\beta(1) = 1$. We compare three paths: *linear* ($\alpha = 1 - t$, $\beta = t$), *trigonometric* ($\alpha = \cos(\pi t/2)$, $\beta = \sin(\pi t/2)$), and a *VP* schedule with linear β_t giving $\alpha^2(t) = \exp(-T(t))$, $T(t) = \frac{1}{2}\beta_{\min} t + \frac{1}{4}(\beta_{\max} - \beta_{\min})t^2$. The training target is the conditional velocity $u_t = \alpha'(t) y_0 + \beta'(t) z$, regressed against the model's $v_\theta(y_t, x, t)$.

Velocity-to-score formula. For any Gaussian path with Wronskian $W(t) = \alpha(t)\beta'(t) - \alpha'(t)\beta(t)$, the conditional score is algebraically recoverable from the population velocity:

$$\nabla \log p_t(y_t \mid x) \;=\; \frac{\alpha'(t)\, y_t - \alpha(t)\, v_\theta(y_t, x, t)}{W(t)\, \beta(t)}. \tag{2}$$

For linear FM this reduces to $-(y_t + (1-t)v)/t$; for trig to $-y_t - (2/\pi)\cot(\pi t/2) v$. We use this identity operationally: it lets a velocity-trained Treeffuser enter the same stochastic-interpolant sampler family as score matching and exposes the endpoint amplification studied in Appendix A.

Stochastic-interpolant family. Albergo et al. [2023] show that any non-negative $\varepsilon(t)$ defines a marginal-preserving reverse-time sampler at the population velocity/score:

$$\mathrm{d}y_\tau = \left(-v_\theta + \tfrac{\varepsilon^2}{2}\, s_\theta\right) \mathrm{d}\tau + \varepsilon\, \mathrm{d}W_\tau, \tag{3}$$

where $\tau = 1 - t$ is reverse time and s_θ is the score from (2). The choice $\varepsilon \equiv 0$ recovers the deterministic probability-flow ODE; positive $\varepsilon(t)$ broadens samples, with exact marginal preservation only for the true velocity/score. We use the schedule $\varepsilon(t) = ct$, which vanishes at the data endpoint where the learned score's $1/\beta(t)$ factor would otherwise dominate (see Proposition 2).

Why these choices suit trees. Three facts guide the implementation (derivations in Appendix A). First, Eq. (2) makes FM compatible with the same stochastic-interpolant samplers as score matching. Second, variance-preserving paths keep the scale of the tree input y_t nearly stationary; a linear path instead has $\mathrm{Var}(y_t) \approx (1-t)^2 + t^2$ after standardization, which compresses the middle of the path and changes split geometry. Third, $\varepsilon = 0$ is a valid member of the sampler family, while positive ε multiplies approximate-score error by the endpoint-sensitive $1/\beta(t)$ factor.

2.3 IMPLEMENTATION NOTES

The flow-matching code reuses Treeffuser's per-output-dimension LightGBM fitter and feature-construction pipeline; only the target and prior change. Sampling uses the same Heun integrator as the score path. The residualizer is unchanged across score and FM variants—a single conditional mean is used regardless of objective. At sample time, samples are inverse-transformed through the residualizer to recover the original y scale. For numerical stability, neither training nor sampling evaluates the singular data endpoint in Eq. (2): score and FM training draw times from $t \geq 10^{-5}$, score sampling reverse-integrates down to 10^{-5}, and FM reverse integration stops at reverse time $1 - 10^{-5}$, i.e. forward path time $t = 10^{-5}$.

3 EXPERIMENTS

Datasets and protocol. Ten UCI benchmarks (Table 2) are evaluated with a nested tuning protocol. We split each dataset into six folds: fold 0 is used only for hyperparameter selection, and folds 1–5 are evaluation folds. For each dataset/model family, Optuna tunes on the fold-0 train/validation split; the selected configuration is then refit and evaluated on each held-out evaluation fold. Evaluation-fold training rows range from 256 on yacht to 38,108 on protein. For sample-based methods, each test point uses 200 samples; Treeffuser sampler choices are detailed in Appendix B. Metrics are averaged over evaluation folds and, for headline summaries, over datasets.

Metrics. We report CRPS skill score $\mathrm{CRPSS} = 1 - \mathrm{CRPS}_{\text{model}}/\mathrm{CRPS}_{\text{climatology}}$, where climatology is the empirical CDF of y_{train}; mean rel-CRPS (per-dataset CRPS divided by the best variant on that dataset, then averaged); the Dawid–Sebastiani score [Dawid, 1984]; the fraction of evaluation folds at which a Kolmogorov–Smirnov test on the probability integral transform fails to reject uniformity at $\alpha = 0.05$ ("PIT KS p>.05"); absolute coverage error at the 50/90/95% nominal interval levels [Gneiting and Raftery, 2007]; and mean sample-generation wall time per benchmark row. We use CRPSS and rel-CRPS for cross-dataset aggregate comparisons because raw CRPS is measured in target units and the benchmark scales vary sharply: mean climatological CRPS across evaluation folds ranges from 0.0043 on `naval` to 44.8 on `diabetes`. Timings are measured on the same Apple-arm64 workstation, exclude fitting, and include generating 200 samples per test point plus inverse transformations. They use practical benchmark settings rather than a single-thread cap: Treeffuser samples in batches with `n_parallel=20` and LightGBM `n_jobs=-1`; QReg-LightGBM and iBUG's XGBoost model use `n_jobs=-1`; CatBoost uses `thread_count=-1`; NGBoost and PyTorch baselines use their library defaults.

Variants compared. Three Treeffuser rows compare bundled recipes: (1) PUBLISHED—the original baseline with noise-prediction, uniform t, no residualization; (2) SCORE+—the baseline plus §2.1; (3) FM—VP flow matching with residualization and a 5-step Heun ODE. Each row is tuned per dataset under the same fold-0 protocol. The residualizer-C configuration used in the headline rows was selected from the prior diagnostic sweep in Appendix D and then fixed; it is not tuned jointly with the diffusion or flow hyperparameters. We also compare against six per-dataset tuned probabilistic baselines: NGBoost [Duan et al., 2020], quantile-regression LightGBM [Ke et al., 2017], CatBoost RMSE-with-uncertainty [Prokhorenkova et al., 2018], iBUG [Brophy and Lowd, 2022], a Gaussian deep ensemble [Lakshminarayanan et al., 2017], and CARD-style neural diffusion [Han et al., 2022].

3.1 HEADLINE RESULT

Table 1 reports cross-dataset means under the tuning/evaluation protocol (DSS is reported in Appendix H because the mean is dominated by a single variance-floor event in iBUG). The score-side recipe lifts CRPSS from 0.670 to 0.681, raises PIT-uniformity pass rate from 0.40 to 0.62, reduces 90/95% coverage error from 0.044/0.023 to 0.020/0.015, and cuts Treeffuser sample time 6.5×. Within this table, flow matching is close on CRPSS, has the best aggregate normalized CRPS among all displayed rows, and is another 2.4× faster than score^{+}, for a 15.4× speedup over the published sampler. The tradeoff is calibration: FM gives back part of score^{+}'s 90/95% coverage advantage. Non-diffusion baselines sample faster still. On raw CRPS, per-dataset winners span five families: a Treeffuser variant wins on 6/10 datasets, while deep ensembles (dia-

Table 1: Headline real-data results, mean over 10 UCI benchmarks with fold 0 used for tuning and folds 1–5 used for evaluation. rel-CRPS is the per-dataset CRPS divided by the best displayed variant on that dataset. $|\text{cE}|@q$ is absolute coverage error at the $q\%$ interval level. The CARD row is a compact benchmark adapter, not a fully optimized CARD pipeline (see Appendix H).

Variant	CRPSS ↑	rel-CRPS ↓	KS $p{>}.05$ ↑	$\|\text{cE}\|$@50 ↓	$\|\text{cE}\|$@90 ↓	$\|\text{cE}\|$@95 ↓	samp. time
Treeffuser-published	0.670	1.242	0.40	0.117	0.044	0.023	221.35 s
Treeffuser-score$^+$ (ours)	**0.681**	1.131	**0.62**	0.069	**0.020**	**0.015**	34.12 s
Treeffuser-FM (ours)	0.680	**1.117**	0.46	**0.064**	0.028	0.024	14.41 s
QReg-LightGBM	0.668	1.459	0.34	0.066	0.041	0.044	0.30 s
CatBoost-unc.	0.638	1.409	0.28	0.093	0.078	0.066	**0.01 s**
NGBoost	0.601	1.550	0.16	0.119	0.114	0.101	0.04 s
Deep ensemble	0.654	1.519	0.26	0.090	0.032	0.018	0.03 s
iBUG	0.649	1.471	0.36	0.111	0.051	0.065	2.89 s
CARD-style diffusion	0.631	1.632	0.24	0.086	0.057	0.043	42.84 s

Table 2: Per-dataset CRPSS (higher is better) and post-split training rows.

Dataset	Train rows	Published	Score$^+$	FM
yacht	256	0.941	**0.962**	**0.962**
diabetes	368	0.201	**0.241**	0.226
energy	640	0.956	0.962	**0.964**
concrete	858	0.740	**0.774**	0.771
wine	5414	**0.346**	0.305	0.307
kin8nm	6826	0.555	0.617	**0.631**
power	7973	0.829	0.839	**0.843**
naval	9945	0.950	0.949	**0.951**
cali. housing	17200	**0.687**	0.676	0.674
protein	38108	**0.494**	0.480	0.472

betes, kin8nm), CARD-style diffusion (naval), and quantile-regression LightGBM (protein) each win at least one. The Treeffuser rows lead on aggregate normalized CRPS and coverage under this protocol, but this aggregate advantage is directional rather than conventionally significant: paired Wilcoxon tests give one-sided $p = 0.080$ for score$^+$ over published and $p = 0.065$ for FM over published; FM versus score$^+$ remains indistinguishable ($p = 0.922$, two-sided).

3.2 PER-DATASET CRPSS

Table 2 shows the Treeffuser per-dataset CRPSS. The improved variants lead on most small-to-mid datasets, but the split is sharper after per-dataset tuning: score$^+$ is best on concrete and diabetes, while FM is best on energy, kin8nm, naval, power, and yacht. The published baseline remains strongest on wine and on the two largest datasets. The resulting message is not that one recipe wins everywhere, but that the score-side modifications and the FM path expose complementary Pareto points: score$^+$ is usually the calibrated choice, FM is the fast normalized-CRPS choice, and the published SDE can remain competitive when the full diffusion sampler is affordable.

Role of ablations. The following ablations are fixed-configuration diagnostics rather than full per-dataset tuned headline comparisons. They are used to isolate sampler and path mechanisms and to document model-selection choices; the confirmatory aggregate comparison is the tuned protocol in Table 1.

3.3 STOCHASTICITY ABLATION

We test 25-step VP-FM SDE variants at matched residualizer-C: a linear schedule with $\varepsilon \in \{0.25, 0.5, 1.0\}$, a $\sqrt{t}$ schedule with $\varepsilon \in \{0.5, 1.0\}$, and a residualizer-E check at $\varepsilon = 1.0$. Across these six SDE settings, no configuration Pareto-dominates the $\varepsilon = 0$ corner (Treeffuser-FM): increasing stochasticity generally worsens CRPSS and coverage error, and SDE sampling requires 25 steps rather than the headline 5-step ODE. This is consistent with Proposition 2: positive ε couples the SDE drift to a $1/\beta(t)$-amplified approximate score, and the ct schedule used here suppresses but does not remove this possible error-amplification mechanism. The closest contender (linear schedule, $\varepsilon = 0.25$) matches $|\text{cE}|$@95 at 0.015 but is worse on CRPSS and $|\text{cE}|$@90; its timing comes from the 25-step SDE diagnostic rather than the full-protein headline protocol (Appendix F.1). This aggregate result should not be read as stochasticity never helping conditionally: the high-IQR diagnostic in Appendix J shows a tail-coverage tradeoff on the largest datasets. The recommendation is therefore deterministic ODE for aggregate CRPS/latency, with stochasticity left as a conditional-calibration knob.

3.4 PATH ABLATION

On an eight-dataset fixed-configuration diagnostic suite, we compare linear, trigonometric, and VP paths with matched residualizer-C and a 5-step deterministic Heun sampler. VP gives the best CRPS, while the two variance-preserving paths reduce 95% coverage error relative to the linear path. The advantage tracks variance preservation (Proposition 1): VP and trig satisfy $\alpha^2 + \beta^2 = 1$, keeping y_t at roughly unit scale across t, which stabilizes the tree regressor's feature space relative to the linear-path schedule. VP-FM CRPS is essentially flat from 5 to 25 Heun steps (4.294, 4.293, 4.300) while sample time grows $\sim 2.7\times$, so five steps is near-optimal rather than a selected latency point.

3.5 NEGATIVE TRANSFERS FROM SCORE TO FM

Three score-side tricks were ported to FM and not carried into the headline recipe after matched sweeps. Log-noise t sampling changes CRPS by $< 0.4\%$ relative to uniform-t; min-SNR loss weighting either matches uniform ($\gamma = 5$) or hurts CRPS ($\gamma = 1$); and mean-scale residualization worsens CRPS by roughly 11–14%. The FM headline therefore keeps the simpler recipe: uniform t, uniform loss weighting, and mean residualization.

4 DISCUSSION

Two findings deserve emphasis. First, the three headline rows compare bundled recipes rather than isolated factors; appendix diagnostics support the proposed mechanisms but are fixed-configuration sweeps, not full per-dataset tuned factorial ablations. Because score$^+$ and FM share the same LightGBM tuning surface, the FM result is not implicitly absorbing tuning-budget improvements that belong to the score track. Second, per-dataset tuning turns the headline into a tradeoff rather than a single-winner story: under the evaluated protocol, score$^+$ is the strongest calibrated Treeffuser row, whereas FM is the strongest normalized-CRPS and latency Treeffuser row. The common mechanism behind both gains is regression conditioning for tree ensembles: EDM preconditioning stabilises the VE score path's inputs and denoising residual, while VP-FM avoids much of the same conditioning problem through the path geometry itself. This is not a single target-scaling story; log-σ sampling controls where tree capacity is spent, the log-σ feature exposes the noise regime, and Heun ODE sampling is an inference-side efficiency gain. The large-dataset inversion argues against a blanket "SDE is worse" interpretation: per-IQR-bin diagnostics show that stochasticity can improve high-uncertainty tail coverage, while the aggregate metrics penalise the same runs for broader intervals, worse CRPS, and higher latency.

Limitations. The comparison is now stronger because all displayed families are tuned per dataset, but it is still finite-budget model selection: the Optuna spaces differ in dimensionality and computational cost. We include NGBoost, iBUG, CatBoost uncertainty, quantile-regression LightGBM, a deep ensemble, and a CARD-style neural diffusion baseline, but not distributional random forests [Cevid et al., 2022]. Headline tables average five evaluation folds after one tuning fold, so per-dataset gaps, especially on the largest datasets, should be read as descriptive rather than as high-powered significance estimates. The remaining modelling limitation is conditional selection: choosing between score$^+$, FM, and stochastic samplers should ultimately be driven by the user's calibration and latency target, not by aggregate CRPS alone.

Future work. Three threads merit investigation: budget-normalized or wall-clock-normalized tuning across probabilistic families, explicit selection criteria for stochastic-interpolant samplers when calibrated *conditional* (rather than marginal) coverage matters, and multi-output regression where the per-output-dimension fitting used here ignores output dependence. Automating residualizer selection via validation CRPS or calibration diagnostics would also remove a fixed design choice that the current protocol inherits from prior sweeps.

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A THEORETICAL DERIVATIONS AND DESIGN CONSEQUENCES

Lemma 1 (Wronskian velocity-to-score identity) *For a differentiable Gaussian path $y_t = \alpha(t)y_0 + \beta(t)z$ with conditional velocity $u_t = \alpha'(t)y_0 + \beta'(t)z$, nonzero Wronskian $W(t) = \alpha(t)\beta'(t) - \alpha'(t)\beta(t)$, and $\beta(t) \neq 0$ (away from the data endpoint), the population velocity $v(y_t, x, t) = \mathbb{E}[u_t \mid y_t, x, t]$ implies the conditional score*

$$\nabla \log p_t(y_t \mid x) = \frac{\alpha'(t)y_t - \alpha(t)v(y_t, x, t)}{W(t)\beta(t)}.$$

Proof. Solving the two linear equations for z gives

$$z = \frac{\alpha(t)u_t - \alpha'(t)y_t}{\alpha(t)\beta'(t) - \alpha'(t)\beta(t)} = \frac{\alpha(t)u_t - \alpha'(t)y_t}{W(t)}.$$

Because $p(y_t \mid y_0, x) = \mathcal{N}(\alpha y_0, \beta^2 I)$, the conditional score is $-z/\beta(t)$. Fisher's identity gives $\nabla \log p_t(y_t \mid x) = \mathbb{E}[-z/\beta(t) \mid y_t, x, t]$. Substituting the expression for z and using that y_t is fixed under the conditional expectation yields the claim. Replacing v by v_θ gives Eq. (2). □

At a fixed t with $\alpha(t) \neq 0$, the same identity inverts to $v(y_t, x, t) = \alpha'(t)y_t/\alpha(t) - W(t)\beta(t)\nabla \log p_t(y_t \mid x)/\alpha(t)$. Thus score and velocity have the same conditional structure as functions of y_t at that time; the practical differences studied here come from path-dependent scale variation across t, feature geometry, and sampler dynamics rather than an intrinsic fixed-t smoothness advantage.

Proposition 1 (Path-induced feature variance) *Let y_0, z be independent with $\mathrm{Var}(y_0) = \mathrm{Var}(z) = I$ after standardization, and let $y_t = \alpha(t)y_0 + \beta(t)z$. Then $\mathrm{Var}(y_t) = \alpha(t)^2 + \beta(t)^2$. For the three paths used in this paper:*

- *Linear ($\alpha = 1 - t$, $\beta = t$): $\mathrm{Var}(y_t) = (1-t)^2 + t^2$, with minimum $1/2$ at $t = 1/2$.*
- *Trigonometric ($\alpha = \cos(\pi t/2)$, $\beta = \sin(\pi t/2)$): $\mathrm{Var}(y_t) = 1$ for all t.*
- *VP ($\alpha^2(t) = e^{-T(t)}$, $\beta^2(t) = 1 - e^{-T(t)}$): $\mathrm{Var}(y_t) = 1$ for all t.*

Proof. Independence and unit variance give $\mathrm{Var}(y_t) = \alpha^2\mathrm{Var}(y_0) + \beta^2\mathrm{Var}(z) = \alpha^2 + \beta^2$. Substituting the three path definitions yields the listed expressions; the trig and VP identities follow from $\cos^2 + \sin^2 = 1$ and from $\alpha^2 + \beta^2 = e^{-T} + (1 - e^{-T}) = 1$. □

Consequence for tree features. Proposition 1 formalises why trig and VP paths keep the tree input near unit scale across the entire path, whereas the linear path compresses to half-variance at $t = 1/2$. For histogram-split trees this matters because the empirical distribution of y_t determines both candidate split locations and where leaf resolution is spent.

Residualization removes predictable location. If $\mu(x) = \mathbb{E}[y \mid x]$ and $r = y - \mu(x)$, then $\mathbb{E}[\|r\|^2 \mid x] = \mathrm{tr}\,\mathrm{Var}(y \mid x) \leq \mathbb{E}[\|y\|^2 \mid x]$. Diffusion or FM on residuals therefore removes a deterministic location component that would otherwise be present at every noise level. The inequality also explains the large-dataset caveat: a fixed residualizer can become a bottleneck once the inner score/velocity model has enough data to represent the mean directly.

Proposition 2 (Endpoint score-error amplification) *Let $v_\theta(y_t, x, t) = v(y_t, x, t) + \delta(y_t, x, t)$ be an approximate velocity with pointwise error δ, and let $\hat{s}_\theta, s$ denote the implied and true scores from Lemma 1. Then*

$$\hat{s}_\theta(y_t, x, t) - s(y_t, x, t) = -\frac{\alpha(t)}{W(t)\,\beta(t)}\,\delta(y_t, x, t),$$

so the pointwise score-error gain is $|\alpha(t)/(W(t)\beta(t))|$. For all three paths considered here $W(t) \to \alpha(0)\beta'(0) > 0$ is bounded near $t = 0$ while $\beta(t) \to 0$, so the gain diverges at the data endpoint.

Proof. Eq. (2) is linear in v, so

$$\hat{s}_\theta - s = \frac{\alpha'(t)y_t - \alpha(t)(v + \delta) - (\alpha'(t)y_t - \alpha(t)v)}{W(t)\beta(t)} = -\frac{\alpha(t)\,\delta}{W(t)\beta(t)}.$$ □

Endpoint amplification. Proposition 2 therefore predicts that any stochasticity schedule with $\varepsilon(t) \not\to 0$ as $t \to 0$ injects high-gain approximate score into the SDE drift; the schedule $\varepsilon(t) = ct$ used in §2.2 and the deterministic $\varepsilon \equiv 0$ corner are the two natural responses.

Proposition 3 (Population equivalence on the VE path) *For a VE path $y_t = y_0 + \beta(t)z$ with $\beta'(t) > 0$, training a normalized velocity model ν_θ against $u_t/\beta'(t) = z$ implies the same population score as VE noise prediction and VE denoising/EDM score reconstruction:*

$$s_t(y_t, x) = -\frac{\mathbb{E}[z \mid y_t, x, t]}{\beta(t)} = \frac{\mathbb{E}[y_0 \mid y_t, x, t] - y_t}{\beta(t)^2}.$$

Proof. For VE, $\alpha = 1$, $\alpha' = 0$, $u_t = \beta'(t)z$, and $W = \beta'(t)$. Lemma 1 gives $s_t = -\mathbb{E}[u_t \mid y_t, x, t]/(\beta(t)\beta'(t)) = -\mathbb{E}[z \mid y_t, x, t]/\beta(t)$. Since $y_t = y_0 + \beta(t)z$, $\mathbb{E}[y_0 \mid y_t, x, t] = y_t - \beta(t)\mathbb{E}[z \mid y_t, x, t]$, which gives the denoising form. Noise prediction estimates $-\mathbb{E}[z \mid y_t, x, t]$ and EDM reconstructs the same population denoiser after its skip/output/input preconditioning. □

Proposition 3 is a population identity, not a claim that the finite LightGBM problems are identical. EDM fits the preconditioned denoising residual $F = (y_0 - c_{\text{skip}} y_t)/c_{\text{out}}$ and reconstructs $D_\theta = c_{\text{skip}} y_t + c_{\text{out}} F_\theta$, whereas normalized VE-FM would fit z directly. They imply the same score at the optimum but present different targets and feature scalings to LightGBM. This is the useful interpretation of the score$^+$–FM tie: both improve the conditioning of the regression problem seen by the tree ensemble, EDM by preconditioning the VE score objective and VP-FM by choosing a better-conditioned probability path.

Table 3: Regression-conditioning view of the main design choices. Here y_0 and z are standardized. "Conditioning issue" refers to the supervised problem fit by LightGBM, not to the population score identity.

Variant	Regressed object	Feature scale	Conditioning issue / fix
VE noise prediction	$-z$	$y_0 + \sigma z$ grows with σ	Unit target, but heteroscedastic input and $1/\sigma$ score amplification.
VE EDM score$^+$	$(y_0 - c_{\text{skip}} y_t)/c_{\text{out}}$	$c_{\text{in}} y_t$ is near unit scale	Input and denoising residual are explicitly preconditioned.
Raw VE-FM	$\beta'(t)z$	$y_0 + \beta z$ grows with β	Velocity target inherits the schedule derivative.
Normalized VE-FM	z	$y_0 + \beta z$ unless also scaled	Population-equivalent to noise prediction after known rescaling.
VP-FM	$\alpha' y_0 + \beta' z$	$\alpha^2 + \beta^2 \approx 1$	Path keeps tree features near the standardized data scale.

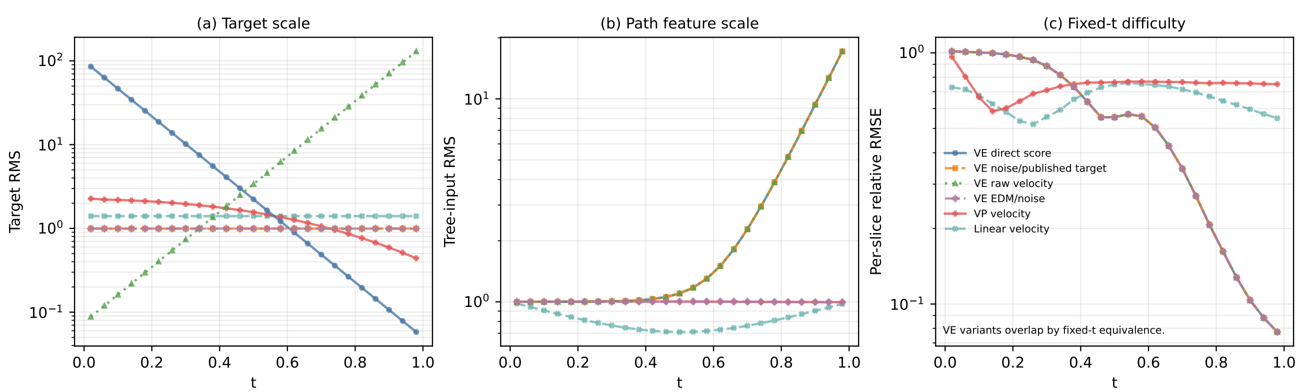

Figure 1: Toy preconditioning diagnostic on a one-dimensional heteroscedastic mixture, generated by `pixi run toy-geometry`. Panel (a) shows that direct VE score prediction and raw VE-FM have target scales that vary by orders of magnitude across t, whereas the published-style VE noise target, EDM/noise prediction, and linear velocity have unit target scale. Panel (b) separates target scaling from input geometry: the published-style VE target is stable but the tree input $y_t = y_0 + \sigma z$ still expands strongly, while EDM rescales the input and VP-FM keeps the path feature near standardized scale. Panel (c) fits a fresh LightGBM at each fixed t; the VE curves overlap up to numerical noise, confirming that the score/velocity distinction is not intrinsic fixed-t smoothness but cross-t conditioning.

B DATASETS AND PROTOCOL DETAILS

Sources. The ten UCI benchmarks follow the canonical Treeffuser protocol [Beltrán-Velez et al., 2024]: `yacht`, `concrete`, `energy`, `wine`, `kin8nm`, `naval`, `power_plant`, `protein` from the UCI repository; `california_housing` from scikit-learn's `fetch_california_housing`; and `diabetes` from scikit-learn's `load_diabetes`. The headline results use the six-fold tuning/evaluation protocol from §3; Table 2 reports the post-split training rows for each evaluation fold. Fixed-configuration appendix diagnostics instead use the train/test sizes declared in their YAML configs, sometimes with large datasets subsampled for sweep cost. The only preprocessing is StandardScaler standardisation of features and target, fit on the training split and applied to the corresponding test split. `wine` is the union of `wine-quality-red` and `wine-quality-white` with a binary `color` indicator.

Seed policy. The fold-tuned headline uses the deterministic six-fold manifest with master seed 0. Fixed diagnostic sweeps use the seed lists declared in their configs (usually 0–2 or 0–9), with three independent streams: `data_seed` (offset 0) for the train/test split; `model_seed` (offset 10,000) for the LightGBM regressors; `sampler_seed` (offset 20,000) for stochastic samplers. Variance reduction across seeds therefore reflects only sampling-and-split variability, not boosting initialisation.

Common training settings. In fixed diagnostic sweeps, unless noted, Treeffuser variants use $n_{\text{estimators}} = 3000$, early stopping after 50 rounds, learning rate 0.1, $n_{\text{repeats}} = 30$ (each training point is reused with 30 fresh t samples), and the per-output-dimension LightGBM fitter. In the headline protocol, these and other LightGBM hyperparameters are selected per dataset from the tuning spaces and stored under `benchmarks/results/tuning/best_params/`. Inference draws 200 samples per test point.

Tuning budget. For each dataset/model-family pair, Optuna targets 25 finite trials on the fold-0 train/validation split, with at most twice as many total attempts to allow failed or non-finite trials. The tuning objective is validation CRPS computed from 100 predictive Monte Carlo samples per validation point; this is the number of generated response samples used to estimate CRPS, not a subsample of validation observations. Final evaluation refits the selected configuration on each held-out evaluation fold and draws 200 samples per test point. The three Treeffuser rows share the same six-dimensional LightGBM search surface: `n_estimators`, `learning_rate`, `num_leaves`, `max_depth`, `min_child_samples`, and `subsample`; method-defining choices such as objective, residualization, probability path, and sampler are fixed by row. External baselines are also tuned for 25 finite trials, but their search spaces and per-trial costs differ, so the comparison should be read as equal finite-trial budget rather than wall-clock-normalized model selection. The persisted YAML files record `n_trials_target`, `n_trials_finite`, and `n_trials_failed`. Table 4 summarises the per-family tunable surfaces.

Appendix calibration metrics. Appendix tables use q-MACE for sample-rank calibration: for each held-out response, we compute the fraction of predictive samples above the observed value, sort these ranks, and average their absolute deviation from an evenly spaced uniform grid. Lower q-MACE therefore means the sample predictive distributions have ranks closer to uniform, analogous to the PIT-KS metric in the main table but reported as an error magnitude.

Sampler details. TREEFFUSER-PUBLISHED uses the VESDE reverse-time SDE with the Euler–Maruyama sampler at $n_{\text{steps}} = 50$, matching the published code path. TREEFFUSER-SCORE^{+} uses the Heun probability-flow ODE at $n_{\text{steps}} = 25$ steps under the EDM-style parametrization with $\sigma_{\min} = 0.01$, $\sigma_{\max} = 20$. TREEFFUSER-FM uses the Heun ODE at $n_{\text{steps}} = 5$ on the VP path with $\beta_{\min} = 0.1$, $\beta_{\max} = 20$.

C SCORE-SIDE MODIFICATIONS: ABLATION CROSS-REFERENCES

Unless otherwise stated, appendix sweeps are fixed-configuration diagnostics used to isolate modelling choices; they are not per-dataset tuned headline comparisons.

The four score-side modifications of §2.1 were validated by smoke and synthetic-core sweeps before being combined in the SCORE^{+} variant. The residualizer-C configuration that the headline uses was selected by the residualizer sweep summarised in Appendix D. Log-sigma time sampling, the log-sigma noise feature, and EDM preconditioning were locked in by the `log_sigma_sweep`, `synthetic_core`, and `pf_ode_sweep` development runs. We treat those runs as model-selection evidence rather than as separately reported ablations; the quantitative claim in the paper is the combined

Table 4: Tunable hyperparameter ranges per model family (25 Optuna trials each). Log = log-uniform sampling. Fixed method-defining parameters are not listed.

Family	Tunable parameter	Range	Scale
Treeffuser (all 3 rows)	`n_estimators`	200–3000	log
	`learning_rate`	0.001–0.3	log
	`num_leaves`	15–255	linear
	`max_depth`	$\{-1, 4, 6, 8, 10\}$	categorical
	`min_child_samples`	5–100	linear
	`subsample`	0.5–1.0	linear
NGBoost	`n_estimators`	200–3000	log
	`learning_rate`	0.001–0.3	log
iBUG	`k`	10–200	log
	`n_estimators`	200–2000	log
	`learning_rate`	0.001–0.3	log
	`max_depth`	3–10	linear
QReg-LightGBM	`quantile_count`	24–49	linear
	`n_estimators`	50–500	log
	`learning_rate`	0.02–0.2	log
	`num_leaves`	15–127	linear
CatBoost	`iterations`	200–3000	log
	`learning_rate`	0.001–0.3	log
	`depth`	4–10	linear
Deep ensemble	`hidden_size`	$\{64, 128, 256, 512\}$	categorical
	`n_layers`	2–5	linear
	`learning_rate`	0.0001–0.01	log
	`max_epochs`	$\{100, 200, 400\}$	categorical
CARD-style	`hidden_size`	$\{64, 128, 256\}$	categorical
	`n_layers`	2–5	linear
	`learning_rate`	0.0001–0.01	log
	`max_epochs`	$\{200, 400\}$	categorical
	`diffusion_epochs`	$\{200, 400\}$	categorical
	`n_steps`	$\{50, 100, 200\}$	categorical

SCORE$^+$ row in Table 1, where all four choices are evaluated together on the full ten-dataset protocol.

D RESIDUALIZER CONFIGURATION

D.1 OFF VS. MEAN RESIDUALIZATION (FM)

We isolate the contribution of mean residualization on the FM side at matched configuration (VP path, uniform t, uniform loss weighting, residualizer-C extra parameters) using a ten-dataset mixed synthetic/real diagnostic suite and ten seeds.

Table 5: FM with mean residualization vs. no residualization (`off`). Paired differences (mean − off, 100 paired observations); negative is better for all metrics shown.

Sampler	ΔCRPS	Δq-MACE	ΔKS stat	Δ |cE|@90	paired t on ΔCRPS
ODE @ 5 steps	-0.23	-0.022	-0.034	-0.032	-4.30
SDE @ 25 steps	-0.05	$+0.001$	$+0.003$	-0.001	-2.91

The ODE-5 advantage is large and broadly significant across calibration metrics (81/100 paired wins on q-MACE and 81/100 on $|\mathrm{cE}|$@90). At SDE-25 the gap on calibration metrics is null (paired $t < 1$); the aggregate CRPS gain is driven by two real-data outliers (`yacht` $+54\%$ relative, `concrete` $+15\%$) while six of ten datasets actually favour the unresidualized baseline. The mechanism is consistent with the few-step ODE being unable to absorb the conditional mean itself when the velocity net has ≤ 5 Heun updates; with 25 SDE updates the velocity model recovers calibration on its own. Cost: mean residualization adds ≈ 12 s of fit time on these sizes ($\sim 8\times$ vs. no residualization), so the trade is most attractive in low-step regimes.

D.2 RESIDUALIZER VARIANTS A–E (FM SIDE)

Five residualizer configurations were swept on a four-dataset, three-seed sub-suite (`california_housing`, `diabetes`, `energy`, `protein`-5k subsample) under VP-FM-ODE at 5 steps; see Table 6.

Table 6: FM residualizer ablation. "Conf. leaves" is the LightGBM `num_leaves`; "$n_{\text{est}}^{(r)}$" is the residualizer's `n_estimators`; "ES" is early-stopping rounds in the residualizer (off means no ES).

Variant	Description	Leaves	$n_{\text{est}}^{(r)}$	ES	CRPSS ↑	q-MACE ↓	|cE|@95 ↓	fit s
A	current-baseline	31	100	off	0.521	0.028	0.017	1.82
B	regularised	31	100	30	0.509	0.042	0.018	2.78
C	high capacity	63	300	off	**0.523**	0.039	0.011	8.03
D	ES moderate	63	500	30	0.516	0.033	0.021	5.91
E	ES + high capacity	63	2000	30	0.522	0.034	**0.010**	9.81

Configurations C and E sit on the calibration–latency Pareto front. C is the headline choice (best CRPSS, $1.2\times$ faster than E); E edges C on tail coverage ($|\text{cE}|$@95 0.010 vs. 0.011). A separate full-protein diagnostic shows E reclaiming the lead on `protein` (CRPSS 0.476 vs. C's 0.449), reinforcing the case that the optimal residualizer is dataset-dependent.

D.3 SCALE-AWARE RESIDUALIZATION

The `mean_scale` residualizer additionally divides residuals by a cross-validated conditional-std estimate $\hat{\sigma}(x)$. On the ten-dataset mixed synthetic/real diagnostic suite with ten seeds:

Table 7: Mean vs. mean-scale residualization on FM at matched configuration. Higher CRPSS is better; lower CRPS, DSS, q-MACE are better.

Variant	CRPSS ↑	CRPS ↓	DSS ↓	q-MACE ↓
VP-FM-ODE, mean	**0.519**	**0.652**	$+0.314$	**0.031**
VP-FM-ODE, mean-scale	0.489	0.724	$+0.665$	0.038
VP-FM-SDE, mean	**0.517**	**0.657**	$+0.380$	**0.040**
VP-FM-SDE, mean-scale	0.475	0.750	$+0.676$	0.049

Scale-aware residualization is uniformly worse: $+11\%$ CRPS on the ODE path, $+14\%$ on the SDE path, with DSS more than doubling. The per-x scale estimator is itself noisy in these finite-sample regimes; dividing by it injects more variance into the velocity target than it removes. `mean` is therefore the only residualizer carried into the headline.

E TIME SAMPLING AND LOSS WEIGHTING

E.1 TIME SAMPLING

Table 8: FM time-sampling sweep on fixed UCI train/test splits, 10 datasets $\times$ 3 seeds, matched residualizer-C, VP path. "uniform" is the headline choice with an endpoint anchor at $t = 1$ at probability 0.05.

Variant	CRPSS ↑	DSS ↓	q-MACE ↓	KS p>.05 ↑	samp s
VP-FM-ODE uniform	**0.6564**	-0.154	0.040	0.47	3.63
VP-FM-ODE uniform anchor16	0.6560	-0.171	0.040	0.53	3.84
VP-FM-ODE logbeta	0.6556	-0.175	**0.035**	**0.77**	3.74
VP-FM-ODE logSNR	0.6538	-0.135	0.042	0.40	3.71
score$^+$ uniform-t	0.6556	-0.210	0.038	0.50	6.95
score$^+$ log-σ t	**0.6565**	-0.169	0.038	**0.70**	6.58

For FM, log-beta and log-SNR sampling do not move CRPS by more than 0.4% relative to uniform with endpoint anchor; the smoother density shape gains $+5\%$ on KS pass rate and -13% on q-MACE but the effect is within seed variance for CRPSS. The headline keeps uniform-t for parsimony; log-beta is a viable substitute when KS uniformity is the priority. For the score model, log-sigma sampling clearly helps PIT uniformity (KS pass $0.50 \rightarrow 0.70$) at the cost of slightly worse DSS; we keep log-sigma in the SCORE$^+$ headline because the KS gain is the more interpretable calibration win.

E.2 LOSS WEIGHTING (MIN-SNR-γ)

Table 9: Min-SNR-γ loss weighting on FM, fixed UCI train/test splits, 10 datasets $\times$ 3 seeds. $\gamma = 5$ collapses to uniform on most of the support; $\gamma = 1$ amplifies the late-time loss most aggressively.

Variant	CRPSS ↑	CRPS ↓	DSS ↓	q-MACE ↓	KS p>.05 ↑
VP-FM-ODE uniform-w	**0.6564**	**4.097**	-0.159	0.040	0.47
VP-FM-ODE min-SNR-5	0.6564	4.119	-0.167	**0.037**	**0.60**
VP-FM-ODE min-SNR-1	0.6516	4.223	-0.104	0.044	0.37
VP-FM-SDE uniform-w	**0.6540**	**4.114**	-0.080	0.051	0.30
VP-FM-SDE min-SNR-5	0.6538	4.138	-0.085	0.050	0.30
VP-FM-SDE min-SNR-1	0.6485	4.239	-0.046	0.057	0.23

Min-SNR-γ is a sharpness-versus-tail trade-off. $\gamma = 1$ hurts CRPS by 0.8–1.3% across both samplers (Wilcoxon $p \leq 0.003$ for both); $\gamma = 5$ is statistically indistinguishable from uniform weighting on CRPS, but improves PIT KS pass rate on the ODE path. We keep uniform weighting in the FM headline.

F STOCHASTICITY (VELOCITY-NOISE) SWEEPS

F.1 STOCHASTICITY STRENGTH

We sweep velocity stochasticity on the VP path under linear and $\sqrt{t}$ schedules with residualizer-C, plus one residualizer-E check; results in Table 10 are averages over fixed UCI train/test splits, 10 datasets $\times$ 3 seeds at 25 SDE steps.

Table 10: Stochasticity sweep. The $\varepsilon = 0$ corner is the deterministic FM recipe (Heun ODE @ 5 steps); SDE rows use 25 steps. Sample time is omitted because the ODE and SDE rows use different benchmark protocols.

Variant	CRPSS ↑	CRPS ↓	|cE|@90 ↓	|cE|@95 ↓
$\varepsilon = 0$ headline (ODE)	0.669	–	**0.024**	**0.017**
$\varepsilon = 0.25$ linear, res-C	0.656	4.099	0.028	0.015
$\varepsilon = 0.50$ linear, res-C	0.655	4.104	0.036	0.017
$\varepsilon = 1.00$ linear, res-C	0.654	4.114	0.047	0.022
$\varepsilon = 0.50$ sqrt, res-C	0.655	4.101	0.035	0.017
$\varepsilon = 1.00$ sqrt, res-C	0.655	4.107	0.043	0.020
$\varepsilon = 1.00$ linear, res-E	0.651	4.128	0.048	0.024

No SDE setting Pareto-dominates $\varepsilon = 0$. The closest contender ($\varepsilon = 0.25$ linear) matches |cE|@95 (0.015 vs. 0.017) but is strictly worse on CRPSS and |cE|@90, and pays 25 SDE steps vs. 5 ODE steps.

F.2 SCHEDULE SHAPE

The schedule $\varepsilon(t) = c\,t$ (linear) is the headline form. The $\sqrt{t}$ alternative was tested for parity at $\varepsilon \in \{0.5, 1.0\}$ and produces statistically indistinguishable CRPSS (Table 10). Both schedules vanish at the data endpoint, which prevents the $1/\beta(t)$ blow-up in the velocity-to-score conversion (Eq. (2)); neither shape helps once ε is small.

Table 11: Flow-path ablation under the 5-step deterministic ODE sampler: linear vs. trigonometric vs. variance-preserving, 8 datasets $\times$ 3 seeds, matched residualizer-C. Lower is better for all columns; bold is best per column.

Path	CRPS ↓	q-MACE ↓	|cE|@95 ↓	samp s
Linear	4.324	**0.026**	0.032	0.49
Trigonometric	4.306	0.028	**0.028**	**0.43**
Variance-preserving (VP)	**4.294**	0.028	**0.028**	0.46

G PATH CHOICE

VP is best on CRPS under the few-step ODE, while trig and VP both reduce 95% coverage error relative to the linear path. The advantage tracks variance preservation: VP and trig satisfy $\alpha^2 + \beta^2 \approx 1$, keeping y_t at roughly unit scale across t and stabilising the tree regressor's feature space. The headline uses VP.

H PROBABILISTIC-FAMILY BASELINES

We compare the headline Treeffuser variants against six external probabilistic regressors (Table 12). Hyperparameters for every displayed family are selected per dataset on the same fold-0 tuning split, then evaluated on folds 1–5. The CARD row uses a compact benchmark adapter with the same two-stage conditional-mean plus conditional-diffusion structure as CARD [Han et al., 2022], so conclusions involving CARD should be read as comparisons to this adapter rather than to a fully optimized CARD pipeline.

Table 12: Probabilistic baselines under the tuning/evaluation protocol. rel-CRPS is normalized by the best displayed variant on each dataset. Sample-time excludes fitting. The iBUG DSS is dominated by a single `wine` fold where the model assigns near-zero predictive variance to a mispredicted test point.

Model	CRPSS ↑	rel-CRPS ↓	CRPS ↓	DSS ↓	q-MACE ↓	|cE|@95 ↓	samp s
Treeffuser-score$^+$ (ours)	**0.681**	1.131	4.057	-0.214	**0.028**	**0.015**	34.12
Treeffuser-FM-fast (ours)	0.680	**1.117**	4.121	$\mathbf{-0.218}$	0.034	0.024	14.41
Treeffuser-published	0.670	1.242	4.279	1.358	0.044	0.023	221.35
Quantile-regression LightGBM	0.668	1.459	4.026	0.016	0.033	0.044	0.30
Deep ensemble	0.654	1.519	**4.008**	0.090	0.050	0.018	0.03
iBUG (XGBoost)	0.649	1.471	4.193	1.85×10^7	0.069	0.065	2.89
CatBoost (uncertainty)	0.638	1.409	4.045	0.272	0.045	0.066	**0.01**
CARD-style diffusion	0.631	1.632	4.838	0.436	0.051	0.043	42.84
NGBoost (Gaussian)	0.601	1.550	4.299	1.957	0.052	0.101	0.04

Under this per-dataset tuning protocol, the Treeffuser variants lead on aggregate CRPSS, rel-CRPS, and coverage metrics. The external baselines are still important: deep ensembles have the best mean raw CRPS, though this scale-sensitive average is less informative about cross-dataset dominance than CRPSS or rel-CRPS, and the per-dataset winners include deep ensembles on `diabetes` and `kin8nm`, CARD on `naval`, and quantile-regression LightGBM on `protein`. Their aggregate rel-CRPS remains worse, indicating that no single tuned external family transfers as evenly across the benchmark suite as the Treeffuser variants.

I EXTENDED PER-DATASET RESULTS

Table 13 reports CRPS, $|\text{cE}|$@90, and sample-generation time on each of the ten benchmarks for the three headline Treeffuser variants, averaged over the five evaluation folds. The "best margin" column gives the gap between the winning variant and the second-best Treeffuser variant in raw CRPS. After per-dataset tuning, the margins are tighter: the main result is no longer a few dramatic wins, but a more balanced Pareto surface between published SDE sampling, score-side conditioning, and FM speed.

FM is the fastest Treeffuser sampler on 9/10 datasets and is especially valuable at scale: on `protein` the speedup is $19.3\times$ over PUBLISHED and $2.0\times$ over SCORE$^+$; on `california_housing` the corresponding speedups are $31.6\times$ and $2.7\times$. On $|\text{cE}|$@90, however, SCORE$^+$ leads on 6/10 datasets and FM on 3/10, which is why Table 1 separates normalized-CRPS and calibration conclusions. Under this evaluation protocol, the per-dataset CRPS ranking reproduces the headline pattern:

Table 13: Extended per-dataset Treeffuser results, ordered roughly by evaluation-fold training size. "best margin" is winner-over-second-best in raw CRPS; the bold cell in each block is the per-metric winner on that row. Numbers come from `benchmarks/results/tuning/eval/*.jsonl`.

		CRPS ↓			$\lvert$cE$\rvert$@90 ↓			sample s		
Dataset	best margin	Pub.	Score$^+$	FM	Pub.	Score$^+$	FM	Pub.	Score$^+$	FM
yacht	FM +0.4%	0.438	0.284	**0.283**	0.092	**0.025**	0.081	0.3	0.3	**0.1**
diabetes	Sco$^+$ +1.9%	35.682	**34.025**	34.660	0.079	**0.029**	0.038	2.4	0.5	**0.2**
energy	FM +5.2%	0.252	0.214	**0.204**	0.081	0.034	**0.022**	2.9	0.8	**0.6**
concrete	Sco$^+$ +1.5%	2.457	**2.126**	2.158	0.034	0.025	**0.023**	3.0	4.8	**0.8**
wine	Pub +5.9%	**0.300**	0.319	0.318	0.037	**0.006**	0.016	100.4	37.0	**11.9**
kin8nm	FM +4.0%	0.067	0.058	**0.055**	0.012	**0.004**	0.023	22.0	20.2	**9.2**
power	FM +2.4%	1.671	1.572	**1.535**	0.013	**0.004**	0.006	14.6	**6.7**	7.6
naval	FM +3.2%	0.000218	0.000220	**0.000211**	0.078	0.065	**0.061**	207.4	91.2	**29.5**
cali. housing	Pub +3.6%	**0.197**	0.204	0.206	0.010	**0.005**	0.007	605.9	52.6	**19.2**
protein	Pub +2.7%	**1.722**	1.769	1.796	**0.005**	0.005	0.007	1254.7	127.0	**64.9**

FM has the best average normalized CRPS, score$^+$ is the more reliable calibration row, and the published SDE remains best on wine, california housing, and protein.

J CONDITIONAL TAIL CALIBRATION DIAGNOSTIC

The large-dataset inversion in Table 2 admits two plausible mechanisms: the fixed residualizer may be suboptimal at high n, and stochastic samplers may improve conditional tail calibration in high-uncertainty regions. Table 14 reports direct 95% coverage on `california_housing` and `protein`, including the highest predicted-IQR bin.

Table 14: High-uncertainty tail diagnostic on the two largest datasets. Cov@95 is marginal empirical coverage; Top-IQR Cov@95 is empirical coverage in the highest predicted-IQR quintile; both target 0.95. IQR-MACE is mean $\lvert$cE$\rvert$@95 across predicted-IQR quintiles, so lower is better. SDE-linear uses a linear stochasticity schedule; SDE-$\sqrt{t}$ uses the square-root schedule. All rows use the full training and test splits in a separate three-seed diagnostic.

Dataset	Variant	CRPS	Cov@95	IQR-MACE	Top-IQR Cov@95
cali. housing	Published SDE	**0.207**	0.958	**0.015**	0.931
	FM ODE-C	0.214	**0.947**	0.021	0.900
	FM ODE-E	0.213	0.947	0.021	0.900
	FM SDE-C linear	0.215	0.969	0.023	0.939
	FM SDE-C $\sqrt{t}$	0.215	0.967	0.022	0.938
	FM SDE-E linear	0.213	0.971	0.023	**0.944**
	FM SDE-E $\sqrt{t}$	0.213	0.969	0.023	**0.944**
protein	Published SDE	1.787	0.944	0.018	0.925
	FM ODE-C	1.860	0.933	0.025	0.923
	FM ODE-E	**1.771**	**0.949**	**0.008**	0.943
	FM SDE-C linear	1.859	0.959	0.011	0.960
	FM SDE-C $\sqrt{t}$	1.856	0.956	0.012	**0.954**
	FM SDE-E linear	1.779	0.974	0.024	0.973
	FM SDE-E $\sqrt{t}$	1.776	0.971	0.021	0.971

The diagnostic supports a tradeoff rather than a one-sided conclusion. On `california_housing`, the deterministic FM rows under-cover the high-IQR bin (0.900), while adding FM stochasticity lifts this coverage to 0.938–0.944; the price is marginal over-coverage (0.967–0.971) and no gain over the published SDE on CRPS or IQR-MACE. On `protein`, residualizer-E with the deterministic ODE is already close to the nominal target and remains best on CRPS and IQR-MACE, whereas the residualizer-E SDE rows over-cover both marginally and in the high-IQR bin. Residualizer-C stochasticity is directionally useful for coverage, especially with the square-root schedule, but gives up CRPS. Thus the safest interpretation is not that SDE samplers are dominated, but that their benefit is conditional and must be selected against conditional-calibration metrics rather than headline CRPS alone.

K REPRODUCIBILITY

The submission includes an anonymized frozen repository via `anonymous.4open.science`. The repository contains the implementation in `src/`, the benchmark harness in `benchmarks/`, the paper sources in `paper/`, and the pixi.toml/pixi.lock environment used for the reported runs. All sweeps in this paper are driven by YAML configs under benchmarks/configs/. Each result file is a JSON Lines document recording the full hyperparameter set, random seeds, the git commit hash of the model code, a hash of the Treeffuser source tree, and every reported metric on a per-dataset/per-fold row. The headline tuning runs persist their selected configurations under benchmarks/results/tuning/best_params/ and their five-fold evaluation rows under benchmarks/results/tuning/eval/. The configs and result artifacts that produced the tables are listed alongside in Table 15; the paper PDF is rebuilt with the paper pixi task.

Table 15: Provenance: appendix tables and the config / result file each was produced from.

Table	Config	Result file
5	fm_off_vs_mean_sweep.yaml	fm_off_vs_mean_sweep__*.jsonl
3	analytic summary	–
6	residualizer_fm_sweep.yaml	residualizer_fm_sweep__*.jsonl
7	fm_mean_scale_sweep.yaml	fm_mean_scale_sweep__*.jsonl
8	fm_t_sampling_sweep.yaml	fm_t_sampling_sweep__*.jsonl
9	fm_loss_weighting_sweep.yaml	fm_loss_weighting_sweep__*.jsonl
10	sde_lower_stoch_sweep.yaml, stochasticity_schedule_sweep.yaml	sde_lower_stoch_sweep__*.jsonl, stochasticity_schedule_sweep__*.jsonl
11	ode_vs_sde_path_ablation.yaml	ode_vs_sde_path_ablation__*.jsonl
12, 2, 13, 1	tuning_manifest.yaml	benchmarks/results/tuning/eval/*.jsonl
14	paper_real_data_v2.yaml, tail_calibration_full.yaml	paper_real_data_v2_full.jsonl, tail_calibration_full.jsonl
4	benchmarks/tuning/search_spaces.py	–